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AGRICONNECT

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DBMS MINI PROJECT
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ABSTRACT

AgriConnect is a web-based agricultural marketplace backed by a PostgreSQL (Supabase) database that connects farmers directly with buyers, reducing dependence on intermediaries. Farmers can publish listings, track stock and manage orders through a status-based lifecycle, with a trigger automatically reducing stock and marking listings sold out. Buyers can browse, search and filter produce and place orders directly. The system records payments, order history, ratings, schemes and market price history, and uses views and stored functions to generate sales, buyer, inventory, rating and revenue reports. A Crop Advisor function matches soil, climate and season inputs against a crop knowledge base, while Aadhaar login stores only a salted SHA-256 hash. The project demonstrates normalization, constraints, triggers, views, stored functions and aggregate queries in a working application.

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INTRODUCTION

AgriConnect is a web-based marketplace that removes middlemen from the agricultural supply chain. Farmers can publish produce and manage orders, while buyers can browse listings, place orders, and track purchases. The system also provides live reports on sales, stock, and earnings, supported by a normalized PostgreSQL database with views and stored procedures, a FastAPI backend, and a Next.js frontend.



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PROBLEM STATEMENT

- Farmers depend on intermediaries, which reduces profit and market reach.
- Produce availability, demand, and pricing are scattered across informal channels.
- Orders and payments are tracked manually, leading to lost records and disputes.
- Buyers lack a reliable way to discover farmers, compare produce, and track orders.
- Sales and inventory data are not captured consistently, so reporting is slow and unreliable.



OBJECTIVES

- Build a direct farmer-to-buyer marketplace that removes intermediaries.
- Enable produce listing, discovery, and ordering in one platform.
- Track every order through a transparent status timeline.
- Recommend suitable crops using soil pH, NPK, moisture, temperature, and season.
- Surface government schemes and subsidies relevant to each farmer.
- Generate reports on sales, active buyers, ratings, inventory, and low stock.

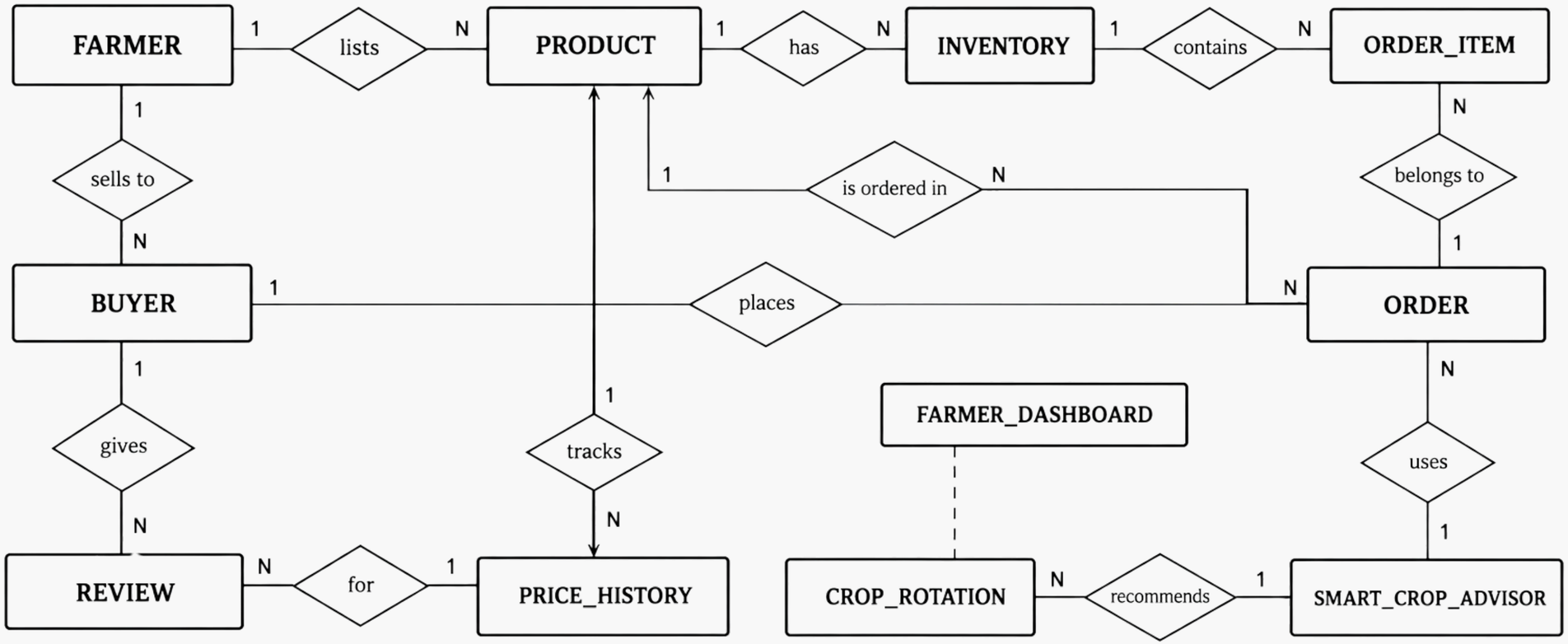


LITERATURE SURVEY

- eNAM (National Agriculture Market): Provides a unified agricultural market but can be complex for small-scale farmers.
- ONDC for Agriculture: Offers opportunities for digital commerce but lacks some integrated advisory and scheme-related features.
- Existing Marketplaces: Primarily focus on e-commerce rather than specialized agricultural workflows.
- Research Gap: A need exists for a simple, role-based platform that combines produce trading, crop advisory, government schemes, and order management.

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ER DIAGRAM



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TECHNOLOGY STACK + SYSTEM ARCHITECTURE

- Frontend: Next.js 16, React 19, TypeScript
- Backend: FastAPI with Uvicorn (Python 3.13)
- Database: Supabase PostgreSQL — normalized schema, views, triggers, stored procedures
- Auth: Aadhaar-based demo login via Supabase RPC (farmer/buyer roles)
- Hosting: Debian 13 NAS + Cloudflare Tunnel for public access

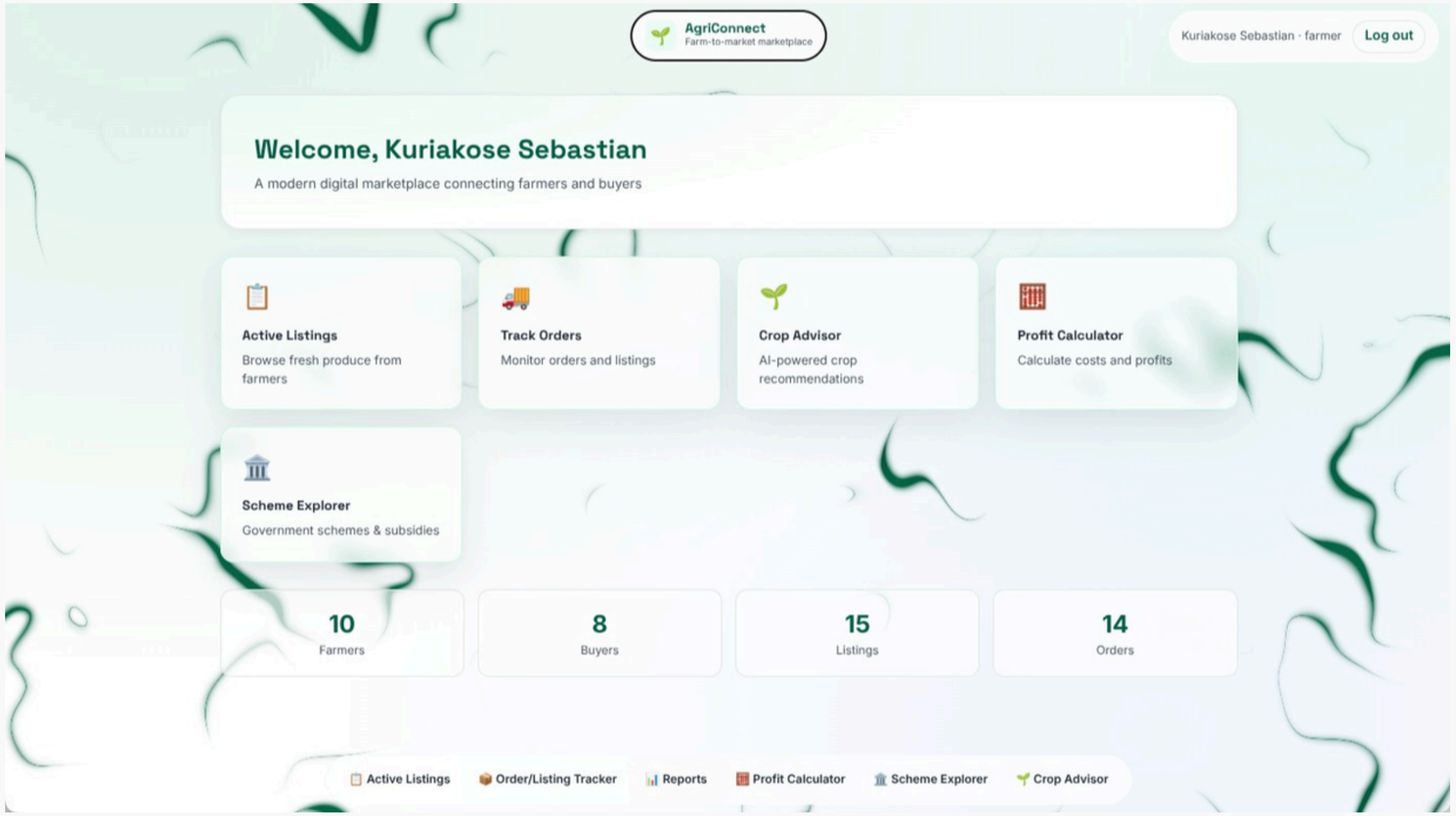


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IMPLEMENTATION & WEBSITE INTERFACE



The AgriConnect website features a clean, modern, and user-friendly dashboard designed to connect farmers and buyers directly. It provides easy access to active listings, order placement and tracking, AI-powered crop recommendations, profit calculation, and government scheme information, while displaying key statistics such as the number of farmers, buyers, listings, and orders for efficient agricultural market management.

MODULES AND FEATURES

- Home Dashboard: Welcome hero, live farmer/buyer/listing/order counts, project overview, feature summary, team, and tech stack.
- Active Listings: Card/table view, search by crop, farmer, or village, and order links for buyers.
- Place Order: Select a listing, enter quantity, validate available stock, and place an order.
- Order Tracker: Track orders through Pending → Confirmed → Delivered → Paid, with cancellation paths, status filters, and All/My orders.
- Crop Advisor: Recommend crops from soil pH, NPK, moisture, temperature, and season using recommend_crops(), with match scores and market outlook.
- Profit Calculator: Estimate revenue, total cost, net profit, margin, and profit per acre from user inputs.
- Scheme Explorer: Show schemes linked to the signed-in farmer, with eligible/ineligible filters and search.
- Reports: Monthly sales, active buyers, farmer ratings, farmer revenue, inventory, and low-stock alerts.
- Login and Authentication: 12-digit Aadhaar-based demo login and registration for farmers and buyers, with guest mode.

DATABASE DESIGN HIGHLIGHTS

Core Tables:

- farmers
- buyers
- listings
- orders
- payments
- ratings
- schemes
- price_history
- crops
- order_status_history

Important Database Views:

- `active_listings_view`
- `order_history_view`
- `order_status_audit_view`
- `inventory_view`
- `low_stock_view`
- `monthly_sales_summary_view`
- `most_active_buyers_view`
- `farmer_avg_rating_view`

Key RPC Functions:

- `aadhaar_farmer_login`
- `aadhaar_buyer_login`
- `recommend_crops`
- `update_order_status`
- `get_farmer_revenue`
- `get_most_active_buyers`
- `get_farmer_ratings`
- `set_farmer_aadhaar`
- `set_buyer_aadhaar`

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UI DESIGN AND IMPLEMENTATION

- Design Theme: Emerald glassmorphism with Space Grotesk/Inter editorial typography.
- Layout: Fixed top dock with brand and account pills, plus a floating bottom navigation dock.
- Visual Design: Translucent glass cards, gradient accents, blurred surfaces, soft green shadows.
- Responsive Design: Desktop-first, mobile-tested at 320, 360, and 390px.
- Consistency: Shared top/bottom docks and role-aware navigation across all modules.



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TESTING AND RESULTS

Testing Performed:

- Role-Based Testing: Verified access restrictions for Farmers, Buyers, and Guests.
- Order Flow Testing: Tested the complete order lifecycle from Pending to Confirmed, Delivered, and Paid.
- Aadhaar Validation: Tested valid and invalid 12-digit inputs.
- Listing and Stock Testing: Verified automatic stock reduction through the database trigger.
- Crop Advisor Testing: Checked crop recommendations using different soil input parameters.
- UI Testing: Tested modules using Streamlit AppTest.

Results:

- All core features function as expected.
- Role-based access restrictions work as intended.
- Stock updates automatically when orders are placed.
- Crop recommendations respond to the provided soil inputs.
- UI testing completed with zero runtime exceptions.

CONCLUSION, FUTURE SCOPE

Conclusion

- AgriConnect bridges the gap between farmers and buyers through a simple digital marketplace.
- Promotes transparency and fairer agricultural trade by enabling direct connections.
- Combines produce listings, order tracking, crop advisory, government schemes, and reports in one platform.
- Supports informed decision-making through data-driven features.

Future Scope

- Multilingual Support: Add Malayalam and other regional languages.
- AI-Based Price Prediction: Forecast crop prices using historical data.
- Online Payment Integration: Integrate UPI or Razorpay for secure transactions.
- SMS/WhatsApp Notifications: Provide real-time order updates.
- Mobile Application: Extend the platform to Android and iOS.
- Logistics Integration: Connect farmers and buyers with transporters for delivery.

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AGRICONNECT AND THE SDG'S

- SDG 2 — Zero Hunger (Targets 2.3, 2.c) - Direct market access and price transparency leads to better incomes for small and marginal farmers.
- SDG 8 — Decent Work and Economic Growth (Target 8.3) - Enables digital entrepreneurship through order management, revenue tracking, and produce listing.
- SDG 9 — Industry, Innovation and Infrastructure (Targets 9.3, 9.c) - A low-cost, database-backed digital marketplace that brings rural enterprises onto modern infrastructure.
- SDG 10 — Reduced Inequalities (Targets 10.1, 10.2) - Reduces information asymmetry and dependence on intermediaries, levelling the playing field for farmers.
- Supporting Goals:
  SDG 1 — No Poverty · SDG 12 — Responsible Consumption · SDG 13 — Climate Action

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